

Gradient LVShim

([Click here for Summary Version](#))

1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Timing Not Applicable	60 mins	Timing Not Applicable

2 Overview

The goal of Gradient LVshim is to improve the shim over a 45-cm (Whole, BRM), 40-cm (ZOOM, CRM) DSV using gradients only.

3 Preliminary Requirements

3.1 Tools and Test Equipment

Item	Qty	Effectivity	Part#	Manufacturer
LVshim Phantom Assembly	1	-	2125245	-
Nesting Plate Assembly	1	-	2125247	-

3.2 Consumables

No consumables required.

3.3 Replacement Parts

No replacement parts required.

3.4 Safety



WARNING

POISON HAZARD!

**1.5T AND 1.0T PHANTOMS CONTAINS NICKEL, A SUSPECT CARCINOGEN.
DO NOT INGEST! DISPOSE OF AS A HAZARDOUS WASTE ACCORDING TO STATE AND
FEDERAL REGULATIONS.**



WARNING

POISON HAZARD!

**3.0T PHANTOMS SILICONE OIL.
DO NOT INGEST! DISPOSE OF AS A HAZARDOUS WASTE ACCORDING TO STATE AND
FEDERAL REGULATIONS.**

3.5 Required Conditions

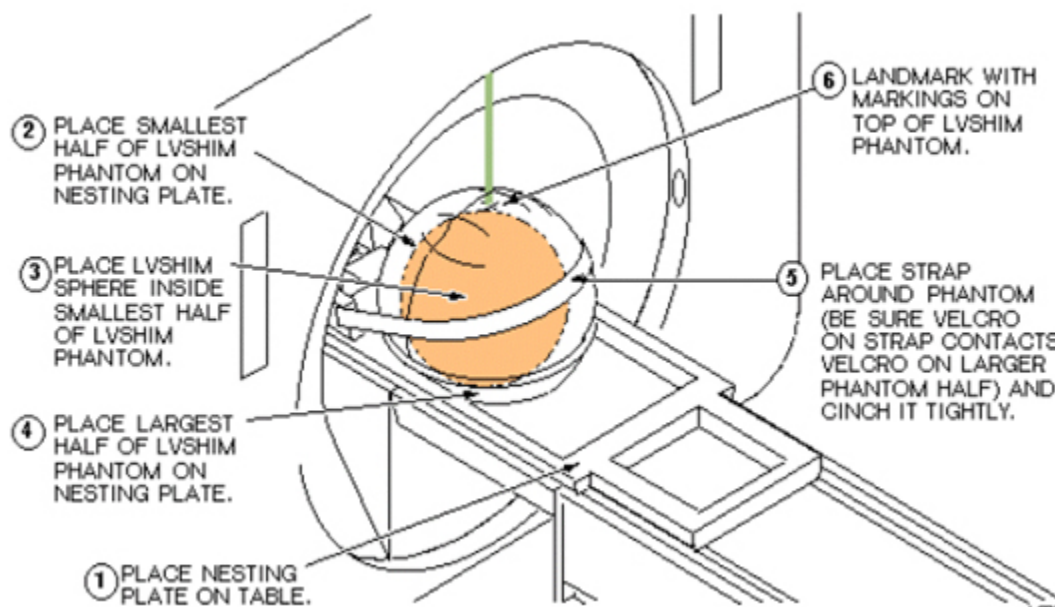
No required conditions.

4 Procedure

4.1 Phantom Setup

1. Remove the Head Coil from the cradle and any phantoms that may be there.
2. Setup and align the LVShim Phantom per [Illustration 1](#).

Illustration 1: Aligning LVShim Phantom on Patient Table



3. Press <Advance to Scan> button.

4.2 Gradient LVShim Scan Prescription

1. At the Host Computer, prepare the system for scan using the [LVShim Scan Protocol](#).
2. Click [Auto Prescan]. Make sure TG is between 130 and 160. Record Gradient Shim current value. If auto prescan fails, proceed to [Step 3](#), (Manual Prescan). Otherwise skip down to [Step 5](#).
3. [Click Manual Prescan]
 1. While in the [Center Freq. Fine (CFH)] mode, verify that the frequency peak is centered in the display. Center frequency peak as necessary. After locating the best signal, open the Frequency menu at the top of Manual Prescan window and select [Save Frequency] to save the new center frequency.
 2. Click [Transmit Gain (TG)], open the Markers menu at the top of the Manual Prescan and select [Horizontal Hairline]. Adjust Transmit Gain for the highest value. TG should be between 130 to 160.

NOTE: The TG needs to be adjusted correctly after performing Auto Prescan. Auto Prescan sets the TG for this phantom too high which results in the RF being over driven and will produce NaN (Not a Number) in the LVShim analysis. Typically the TG will maximize at approximately 20 counts less than the Auto Prescan setting when adjusted in Manual Prescan.

3. Click [Scan TR (R1/R2)]. Verify that R1 and R2 are not over ranged. Adjust R1 and R2 for approximately 50%.
4. Click [Done] to exit Manual Prescan.
5. When finished with manual prescan, click [Scan].

NOTE: The annotation on the LVShim images is not correct and does not change. The scan plane is driven by the LVShim PSD, not through scan prescription. Therefore, the software that is responsible for the image annotation doesn't know about the plane changes and keeps the annotation at the first plane. The images are in the correct plane and orientation; however, the annotation does not reflect this.

4.3 Gradient LVShim Analysis

1. At the Host Computer, go to the Service Desktop and open the Service Browser, if it's not already running.
2. Start the LVShim tool as shown in [Illustration 2](#).

Illustration 2: Starting LVShim Tool



3. For TwinSpeed, highlight WHOLE or ZOOM matching the coil currently being shimmed.
4. The LVShim tool will started as shown in [Illustration 3](#).

Illustration 3: LVShim Main Menu

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1. Shim Type (Gradient, Sup18 1.5T, TEST): Sup18 1.5T
2. Image Data (Ex, Se, Im Number)      : 5007, 3, 1
3. Operation Mode (Service or Research) : Research
4. Display Mag and Phase Diff Images    : No
5. Calibration File Name : lv_15t_18sc_45dsv.full
6. Existing Current in Each Coil.

    AX_1:  4.704    T1_1:  6.433    T2_1: -2.557
    AX_2: -0.702    T1_2:  2.892    T2_2: -8.939
    AX_3: -6.360    T1_3: -7.267    T2_3: -0.203
    AX_4: -6.531    T1_4:  2.926    T2_4: -2.769
    AX_5: -0.364    T1_5: -1.048    T2_5: -0.039
    AX_6: -9.274    T1_6:  0.485    T2_6: -2.158

0. Accept (q or s to quit).

```

Enter the Index Number to Change Default (0..6) [0]:

5. Change the following parts to calculate the next set of currents while performing Main LV Shim:
 - a. Shim Type should be set to **Gradient** . Type **1** < Enter> until the correct set is displayed.
 - b. Image Data should show the Exam, Series and the first Image number of the previously scanned LV Shim series. Type **2** < Enter> and change if necessary.
 - c. Operation Mode should be set to Service. Type **3** < Enter> until this is listed correctly.
 - d. Display Mag and Phase Diff Images should be set to **No**. Type **4** < Enter> until this is listed correctly.
 - e. The calibration file should list the correct configuration file to use. Type **5** < Enter> to view possible files. You can view detailed information on these files in the [LV Shim Software](#) document. Make sure the correct gradient mode is chosen if this system is a TwinSpeed.
 - f. The existing grad shim in each coil should match the grad shim set in each coil from the previous iteration (or the initial currents). Type **6** <Enter> and modify the currents if necessary.
6. Once the LVShim Analysis tool is setup correctly via [Step 5](#), type **0** <Enter> to run the analysis.
7. If image phase wrap was detected, the following message appears: "Excessive phase wrapping....Please re-scan at Wider Bandwidth or smaller test diameter". If this occurs, return to [Gradient LVShim Scan Prescription](#) and rescan and a greater bandwidth.
8. The output should open in a c-shell. See [Illustration 4](#) for an example. (Bandwidth, Exam, Series and Image # will match the LV Shim series under analysis):

Illustration 4: Example Output

PLANE	MIN	MAX	P-P	AVE	STD	1.	-114	50	164	-40.00	26.51
2.	-132	44	176	-39.33	25.31						
3.	-138	40	178	-39.91	32.48						
4.	-138	32	190	-50.04	36.14						
5.	-140	46	186	-48.49	34.68						
6.	-122	48	170	-48.76	32.53						

Exam 50007, Series 3, Image 1											
6 Planes, Bandwidth = 200 Hz, FOV = 60 cm, DSV = 45 cm											
Inhomogeneity = 31.98 Hz. (0.504 ppm)											
Harmonic Coefficients (Zero to Peak)											
Z1:	-8.19	X:	-8.70	Z2X:	-3.44						
Z2:	4.55	Y:	36.93	Z2Y:	-0.49						
Z3:	3.71	ZX:	-5.56	ZX2_ZY2:	0.64						
Z4:	0.08	ZY:	4.76	ZXY:	2.83						
Z5:	-2.64	X2_Y2:	2.12	X3:	-3.73						
Z6:	5.25	XY:	-2.36	Y3:	-1.83						

Generating LVshim report file											
=====											
Please enter two lines of comments (Max 70 characters/line)											
Press Return at the end of each line											
To leave a comment line empty, press Return											
----- Max comment length -----											

9. If "NaN" is displayed in the Harmonic Coefficient data, you should return to [Gradient LVShim Scan Prescription](#) and rescan. Make sure the TG value is between 130 and 160.
10. Enter Comments as desired. Once comments are entered, the following is displayed :

Coil	Existing	Delta	New
X	1	-1	0
Y	0	0	0
Z	3	1	4
11. Record this data in the appropriate [Data Sheet](#).
12. Compare this data to the appropriate LV Shim Specifications for your magnet, see [CRM TRMZM \(Zoom\)](#) or [BRM TRMWB \(Whole Body\)](#). Keep iterating until no noticeable shim improvement is seen.
13. If you accept the new grad shim values, type Y <Enter> and return to [Gradient LVShim Scan Prescription](#) and rescan, the new currents are automatically entered into the gradient amplifiers by the software. Update the [Data Sheets](#).

5 Finalization

1. Remove any phantoms and or tool used during LVShim.
2. Do a [TPS Reset].
3. Do a head or body test scan before customer turn over.

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